Unsupervised Joint Adversarial Domain Adaptation for Cross-Scene Hyperspectral Image Classification

Xuebin Tang, Chunchao Li, and Yuanxi Peng

***Abstract*— In practical hyperspectral image cross-scene classification (HSICC) tasks, the arduous work of obtaining labels and the distribution inconsistency caused by spectral shift leave deep learning methods to face great challenges. Unsupervised domain adaptation aims to exploit knowledge from the annotated source domain and transfer it to the unlabeled target domain, thereby boosting the performance of unsupervised classification. Nevertheless, existing HSICC approaches cannot effectively exploit class structure information from target data. In particular, this article proposes an unsupervised joint adversarial domain adaptation (UJADA) architecture for HSICC to further narrow the distribution gap between distinct domains. The proposed method contains two modules: domain adversarial module that learns domain-invariant features, biclassifier adversarial module that explores task-specific decision boundaries between classes, and both share a feature generator consisting of the dense-based spectral–spatial convolution network. The UJADA simultaneously considers domain- and class-level feature alignment between source and target hyperspectral images in unified adversarial learning processing. Furthermore, the classifier determinacy disparity metric is introduced to fine-grained measure the output probabilistic discrepancy between two task-specific label predictors on target data, thus ensuring the discriminability of transferable features. Comprehensive experiments and ablation studies conducted on two public cross-scene data pairs and our newly acquired ultralow-altitude hyperspectral images under different illumination conditions demonstrate the superior performance of the proposed algorithm, which will greatly promote the practical application of hyperspectral intelligent perception technology.**

***Index Terms*— Classifier determinacy disparity (CDD), hyperspectral image cross-scene classification (HSICC), joint adversarial, unsupervised domain adaptation (UDA).**

I. Introduction

HYPERSPECTRAL image (HSI) provides abundant information combined with spatial texture and hundreds of continuous and subdivided spectral bands, which is beneficial to distinguish different materials in the objective imaged scene. Therefore, HSI has attracted extensive attention from numerous domains, such as medical imaging, geological prospecting, environmental monitoring, and crop yield estimation [1]. Pixel classification for land cover is a crucial step in lots of applications of hyperspectral remote sensing. Over the past decade, a plenty of approaches have been proposed to improve the HSI classification performance via exploiting spectral information assisted with spatial features [2]. Compared with traditional hyperspectral classification methods that relied on handcrafted features [3], [4], [5], [6], deep neural networks (DNNs) have attracted enormous attention in hyperspectral image classification for years and made breakthroughs, benefiting from their advantages in automatically extracting features with representative and discriminative information [7], [8], [9], [10], [11], [12], [13].

While most existing deep learning models primarily focus on HSI classification in a supervised manner, as we know, training supervised DNNs generally requires sufficient labeled samples. Despite their tremendous success, it is strenuous and not practical to acquire training samples with high quality for all remote sensing images, which causes a great challenge in hyperspectral image classification [3], [7]. In particular, in the hyperspectral image cross-scene classification (HSICC) task, which is more related to real-world applications, the source scene and the target scene share the same categories, and the same kind of material in different scenes are likely to have similar properties. A natural thought is to mine valid information only delivered from labeled samples in the source domain and exploit it to improve the classification performance on unlabeled samples in the target domain. However, due to the variability of the external environment, such as atmospheric and illumination conditions, the acquisition angle of the sun, the adjacency effect, and the variation in physical characteristics of the material over time, there is usually a phenomenon of spectral shift between hyperspectral images in different scenes [14], [15], [16], which causes pixels belonging to the same land cover class to possess different spectral signatures. This conflicts with the distributional consistency assumption that machine learning follows. Thus, since the distribution disparity between images is completely ignored when transferring the model trained on source data to target data directly, the classification performance will attenuate sharply.

To mitigate the domain shift, unsupervised domain adaptation (UDA), a major branch of transfer learning, emerges as the times require, which has been proven to realize distribution alignment across domains and effectively improve the performance on an unlabeled target domain by utilizing shared knowledge with a related labeled source domain. Over the past few years, UDA has shown a promising effect in the remote sensing community. In this regard, many shallow-structure

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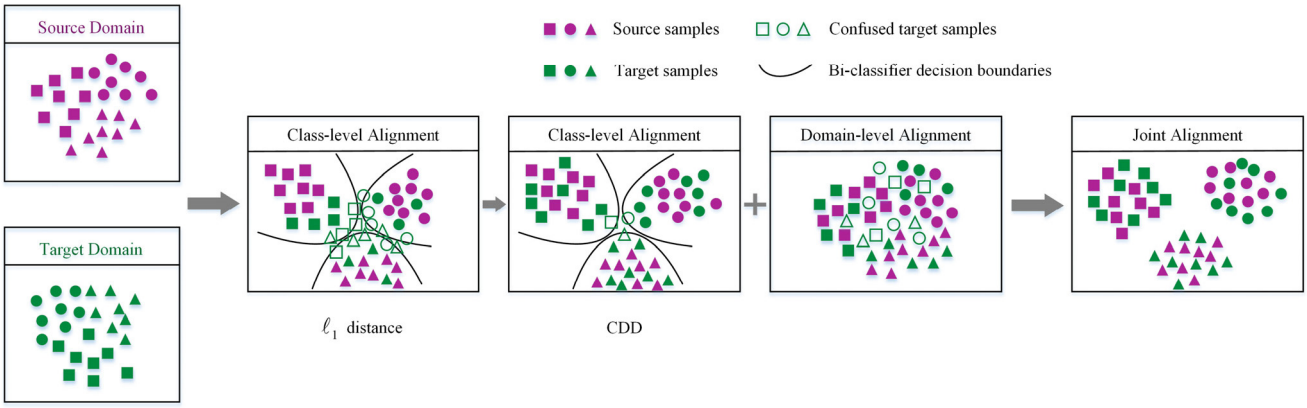


Fig. 1. Motivation of the proposed UJADA approach, the ℓ_1 distance, and the CDD are two different metrics of biclassifier output discrepancy. We adopt the CDD metric to improve the vague classification boundary of target samples caused by the ℓ_1 distance.

domain adaptation algorithms have been proposed, e.g., methods for reweighting data instances [17], [18] and shared feature representation [14], [16], [19], [20]. Notably, end-to-end deep domain adaptation methods become prevailing due to their extraordinary ability to migrate more effective information by embedding adaptation modules in neural network architecture [21], [22], [23]. A common paradigm in UDA aims at decreasing the distribution divergence across source and target domains by minimizing some specific distance metrics [24], [25], [26], [27], [28], [29], [30] such as maximum mean discrepancy (MMD) [27] and correlation alignment (CORAL) [28].

Besides, another paradigm leveraging adversarial learning strategy [31] has also achieved exceptional success by learning domain-invariant representations in a two-player game [32], [33], [34], [35], [36], [37]. There exist two different routes to implement adversarial domain adaptation either by applying an extra domain discriminator to distinguish which domain a sample belongs to [32], [33], and [34] or by adopting a within-network adversarial strategy with two distinct label predictors to draw task-specific decision boundaries between classes [35], [36], [37]. Regarding the first fashion, i.e., domain-level alignment, the domain adversarial training of neural networks (DANNs) embeds a gradient reversal layer in backpropagation training for feature alignment [32], which ignores the category information of target data and may bring side effects of the deterioration of the feature discriminability. The adversarial discriminative domain adaptation (ADDA) model combines discriminative modeling, untied weight sharing, and a generative adversarial network (GAN) loss to solve domain drift [33]. The idea of ADDA has also been extended to HSICC and gained satisfactory classification performance [38], [39], [40], [41]. In particular, Yu *et al.* [40] recently proposed two UDA models for the HSICC task successively, namely, UDA with dense-based compaction (UDAD) [40] and UDA with content-wise alignment (UDACA) [41]. The UDAD is the first attempt to implement UDA for HSI with dense feature compaction, which extracts discriminative feature representation in the annotated supervised training phase. The UDACA innovatively exploits better invariant feature representation for cross-scene hyperspectral images by performing the content-wise alignment. However, these domain-wise UDA methods may easily result in the mode collapse eventually in specific classes with apparent discrepancy if they are not optimized appropriately.

As for the biclassifier mode, the maximum classifier discrepancy (MCD) approach [35] can detect target samples that are far from the source support effectively through the minimax game on cross-classifier outputs disparity between the feature generator and two task-specific classifiers. Therefore, the choice of classifier discrepancy metric is essential for expected adaptability. Li *et al.* [42] devised a classifier determinacy disparity (CDD) metric in the bi-classifier determinacy maximization (BCDM) framework to facilitate the generation of discriminative representations by considering the classifier determinacy in the target domain while utilizing the biclassifier disagreement. However, since class-level alignment methods heavily depend on the accuracy of source-supported predictors, their capability will decay drastically once domain shift is tremendously large [43], which is common in reality HSICC tasks. In the field of daily RGB image recognition, Li *et al.* [44] developed a joint adversarial domain adaptation (JADA) approach to align domain-level and class-level distributions simultaneously. However, the class-level alignment in the JADA framework only utilizes the ℓ_1 distance to measure the output difference of the two task-specific classifiers in a coarse-grained manner, making the classifiers less robust to perturbations.

To solve the aforementioned problems, this article presents a solution to the HSICC task based on a novel unsupervised JADA (UJADA) framework. Fig. 1 renders the motivation of UJADA. The proposed method can further reduce the distribution gap caused by the spectral shift between different hyperspectral images through integrating domain adversarial module and biclassifier adversarial module. Moreover, CDD metric [42] and dense-based spectral–spatial convolution network [45] are deployed to ensure that the learned transferable features will possess discriminable structures and obtained performance improvements. In this way, cross-domain knowledge transfer can be carried out efficiently by mitigating the class-level mismatching and discriminative information missing when annotated information of the target samples is not available. The major contributions of this article can be summarized as follows.

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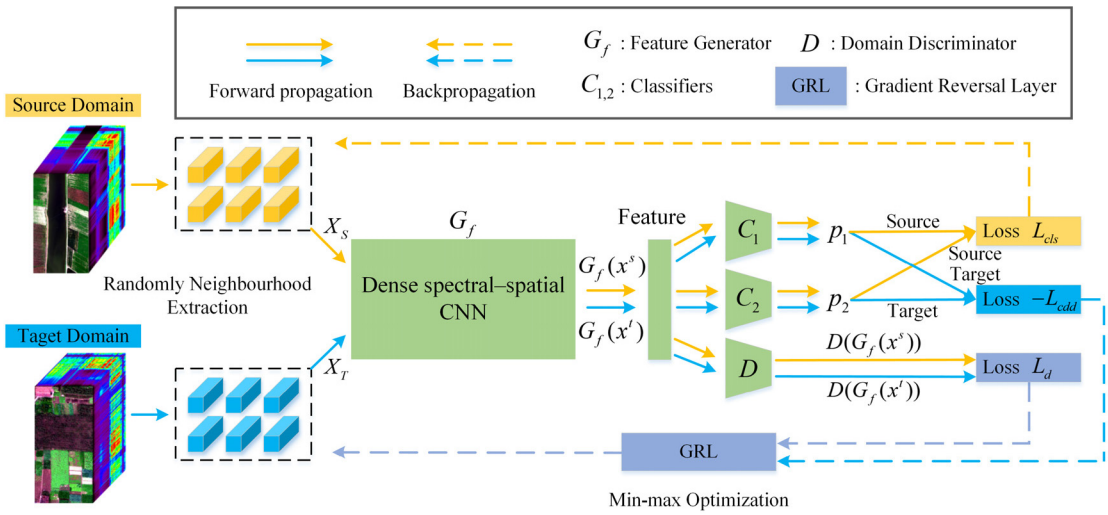


Fig. 2. Overall architecture of UJADA.

1) We propose a UJADA architecture for the HSICC task, which enables efficient feature alignment via end-to-end adversarial training. To the best of our knowledge, this is the first attempt at applying the JADA approach to the hyperspectral remote sensing field.
2) We employ the CDD metric to measure the output discrepancy of two task-specific classifiers during joint adversarial training with the assistance of a dense-based spectral–spatial convolution network, facilitating the generation of domain-invariant representations and discriminative target features.
3) We conduct extensive crossover experiments on three data pairs with different domain shifts, containing multitemporal images and images of spatially disjoint geographical areas and verifying the excellent classification performance of UJADA. It is worth noting that self-collected ultralow-altitude hyperspectral images acquired by an unmanned aerial vehicle (UAV) under different lighting conditions are utilized to demonstrate the practical value of the proposed method for the first time.

The remainder of this article is organized as follows. Section II describes the architecture and training strategy of the proposed UJADA model. Section III provides the details of experiments with the corresponding analysis. The conclusion is drawn in Section IV.

II. Proposed Approach

The proposed UJADA is devised to explore cross-domain knowledge for hyperspectral image classification, which not only has the basic capability to extract spatial–spectral features but also can guarantee the domain invariance and distinguishability of learned features under the cross-scenario situation. In the rest of this section, we describe the related components in detail.

A. Problem Setting

In the setting of UDA, we denote the labeled source domain as $X_S = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ with n_s samples, where y_i^s is the corresponding label of x_i^s. Similarly, the unlabeled target domain with n_t samples is denoted by $X_T = \{x_i^t\}_{i=1}^{n_t}$, where $x_i^s, x_i^t \in \mathbb{R}^{H\times W\times B}$ and $y_i^s \in \{0, 1, 2, \ldots, K\}^{H\times W}$ (H and W are the height and width of the image, respectively, B is the number of spectral bands, and K is the number of ground-truth classes).

B. Overall Network Architecture

The overview of the proposed UJADA is presented in Fig. 2, including a feature generator with the fast dense spectral–spatial convolution (FDSSC) network framework [45], a domain discriminator, and two classifiers. The three parts work in a joint adversarial manner to comprehensively align the domain-wise distribution between source and target domains as well as class-wise.

Specifically, we adopt the architecture of FDSSC as the feature generator, which is composed of dense blocks as the backbone. Thus, the spectral–spatial joint features can be extracted for the cross-scene classification task, where the learned spatial information and spectral knowledge are complementary to each other during training. Moreover, hyperspectral image patches were used as input samples, which are cropped around pixels being processed. Therefore, we can perform more discriminative spectral–spatial representations under supervised training in the source domain and implement efficient alignment of spectral–spatial features in unsupervised joint adversarial training. In addition, both the domain discriminator and classifiers consist of two fully connected layers with a dropout rate of 0.5. The activation function of the first fully connected layer is leakyReLu [46], and the second layer outputs the softmax probabilities [47].

C. Domain-Level Distribution Alignment

Domain adversarial training of neural networks (DANN) has been proven to be effective in solving deep domain adaptation problems [48], as well as the HSI cross-scene classification problem. In particular, two task-specific label classifiers exist in the proposed model for subsequent class-level alignment.

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As shown in Fig. 2, during the domain adversarial training phase, we first aim to minimize the label prediction loss on the annotated HSI cubes of the source domain so that the parameters of both the feature generator and two task-specific label predictors can be optimized to reduce the empirical loss for source data. This guarantees the identifiability of the feature $G_f(x^s)$, which can be formulated as follows:

$$\min_{\theta_g,\theta_{c_1},\theta_{c_2}} L_{\text{cls}}\left(x_i^s, y_i^s\right) = \frac{1}{2n_s}\sum_{i=1}^{n_s}\sum_{j=1}^{2} L_{ce}\left(C_j\left(G_f\left(x_i^s\right)\right), y_i^s\right) \tag{1}$$

where L_{ce} represents the cross-entropy loss we adopt. In order to achieve domain-level alignment and learn global domain-invariant features, the domain discriminator follows the feature generator and is dedicated to accurately distinguishing samples from the source domain or the target domain. Specifically, we define the domain labels of the source domain and target domain as 1 and 0, respectively. Thus, the parameters θ_d of the domain discriminator D are optimized by minimizing the domain discrimination loss

$$\min_{\theta_d} L_d\left(x_i^{s,t}, d_i^{s,t}\right) = \frac{1}{n_s + n_t}\sum_{x_i \in (X_S \cup X_T)} L_{ce}(D(G_f(x_i)), d_i). \tag{2}$$

On the contrary, the feature generator G_f is trained to confuse D simultaneously. The parameters θ_g of G_f are updated by maximizing the loss of domain discriminator

$$\min_{\theta g} -L_d\left(x_i^{s,t}, d_i^{s,t}\right). \tag{3}$$

Accordingly, the optimal parameters $\hat{\theta}_g$, $\hat{\theta}_{c1}$, $\hat{\theta}_{c2}$, and $\hat{\theta}_d$ can be obtained after the training convergence as follows:

$$\left(\hat{\theta}_g, \hat{\theta}_{c1}, \hat{\theta}_{c2}\right) = \arg\min_{\theta_g,\theta_{c_1},\theta_{c_2}} L_{\text{cls}}\left(x_i^s, y_i^s\right) - \beta L_d\left(x_i^{s,t}, d_i^{s,t}\right) \tag{4}$$

$$\left(\hat{\theta}_d\right) = \arg\min_{\theta_d} \beta L_d\left(x_i^{s,t}, d_i^{s,t}\right). \tag{5}$$

Here, β is the tradeoff hyperparameter to balance the label classification loss L_{cls} and the domain prediction loss L_d.

D. Class-Level Distribution Alignment

In practical HSI cross-scene classification tasks, it is very likely to confuse source and target data in the learned feature space when only performing the domain-level alignment, especially for data close to the decision boundary of the HSI classification task. To achieve better feature alignment, it is necessary to preserve the discriminative information of the categories while matching the source and target distributions. Inspired by MCD [35] and JADA [44], we observe that the class structure information provided by the classification boundaries of two task-specific source classifiers can be effectively exploited to predict target domain samples near the decision boundary. In this way, the feature generator is forced to generate features of target samples near the support of the source, thereby facilitating class-level matches between domains. We denote p_1 and p_2 as the output softmax probabilities of two task-specific classifiers. It is obvious that p_1 and p_2 are given as follows:

$$\sum_{k=1}^{K} p_j^k = 1, \quad \text{s.t. } p_j^k \geq 0 \quad \forall k = 1, \ldots, K, \;\; j = 1, 2 \tag{6}$$

where p_j^k represents the kth element of the predicted probability vector of C_j and K indicates the number of classes. However, previous methods [44] adopt the ℓ_1 distance $|p_1 - p_2|_1$ to measure the distribution divergence between two classifiers' predictions, which only considers the similarity of prediction vectors at the corresponding category (i.e., the same position of the vector), thus ignoring the intercategory correlations. This may lead to lower confidence in predictions. Therefore, we exploit the CDD metric [42] in JADA for HSI cross-scene classification, which ensures the diversity of classifiers' output probabilities and the discriminability of the generated features. Given H as the biclassifier prediction relevance matrix

$$H = p_1 p_2^{\text{T}} \tag{7}$$

where H is a square matrix with size $K \times K$ and $H_{mn} = p_1^m p_2^n (m, n \in \{1, 2, \ldots, K\})$ is the element in the mth row and nth column of H. H could effectively assess the biclassifier prediction relevance across different categories. Therefore, we make the two predictions consistent and correlated by minimizing the classifier discrepancy with respect to the prediction relevance, i.e., maximizing the diagonal elements of H. Conversely, the nondiagonal elements of H can be considered as fine-grained confusion information between the two classifiers. Then, the CDD loss is calculated as

$$\Gamma(p_1, p_2) = \sum_{m,n=1}^{K} H_{mn} - \sum_{m=1}^{K} H_{mm} = \sum_{m \neq n}^{K} H_{mn}. \tag{8}$$

It can be seen that CDD measures the discrepancy of distinct predictions by accumulating the product of inconsistent elements in two probability vectors output from C_1 and C_2. Specifically, CDD is defined as the summation of all the probabilities that the biclassifier predictions are inconsistent. To learn transferable features and distinguishable decision boundaries on the target domain, we first train two different label predictors C_1 and C_2 to maximize CDD loss on target data while preserving the supervision from source samples

$$\min_{\theta_{c_1},\theta_{c_2}} L_{\text{cls}}\left(x_i^s, y_i^s\right) - \beta L_{\text{cdd}}\left(x_i^t\right) \tag{9}$$

$$L_{\text{cdd}}\left(x_i^t\right) = \frac{1}{n_t}\sum_{i=1}^{n_t} \Gamma(p_{1i}, p_{2i}) \tag{10}$$

which enforces classifiers to produce disagreement across categories, thus exploring diverse output spaces under the source support. From a mathematical point of view, the CDD loss efficiently promotes two classifiers to perform variously across categories rather than ℓ_1 distance, which potentially encourages the whole G_f, C_1, and C_2 to detect more ambiguous target samples near the decision boundary. To achieve class-level adversarial alignment, we proceed to train the feature generator G_f on target data to minimize CDD loss, so as to push the two different domains together class-wisely

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by facilitating generating discriminative representations that make the predictions determined and consistent

$$\min_{\theta_g} \beta L_{\text{cdd}}\left(x_i^t\right). \tag{11}$$

Consequently, the prediction consistency ensures the target samples to get close to the source spaces class-wisely, which further improves the accuracy.

To sum up, we simultaneously conduct two minmax optimizations by comprehensively aligning both domain- and class-level distributions via utilizing a gradient reversal layer. For brevity, we summarize the training procedure of UJADA based on the aforementioned elaboration as in Algorithm 1. Finally, the classification performance of the proposed algorithm is tested on target datasets using the trained feature generator and classifiers.

Algorithm 1 Training Procedure for UJADA

Input: The set of labeled source samples $\left\{\left(x_i^s, y_i^s\right)\right\}_{i=1}^{n_s}$ and the set of unlabeled target samples $\left\{x_i^t\right\}_{i=1}^{n_t}$, the randomly initialized generator G_f, domain discriminator D and classifiers C_1 and C_2, Trade-off parameter: β; Batch size: M, the maximal epoch number N.

Output: Optimal parameters $\hat{\theta}_g$, $\hat{\theta}_{c1}$, $\hat{\theta}_{c2}$ of feature generator G_f and classifiers C_1, C_2.

For epoch $\leftarrow$1 to N **do**

Step 1: Randomly sample mini-batch of M source samples and M target samples.

Step 2: Train G_f, C_1 and C_2 under the source supervision, as well as learn global domain-invariant features on both source and target data by maximizing the domain adversarial Loss:

$$\min_{\theta_g,\theta_{c_1},\theta_{c_2}} L_{cls}(x_i^s, y_i^s) - \beta L_d(x_i^{s,t}, d_i^{s,t}).$$

Step 3: Train D to achieve domain-level adversarial learning by minimizing the domain adversarial Loss:

$$\min_{\theta_d} \beta L_d(x_i^{s,t}, d_i^{s,t}).$$

Step 4: Update $\theta_{c_1}, \theta_{c_2}$ by maximizing classifier determinacy disparity and simultaneously maintain the accuracy on source samples:

$$\min_{\theta_{c_1},\theta_{c_2}} L_{cls}(x_i^s, y_i^s) - \beta L_{cdd}(x_i^t).$$

Step 5: Train G_f to realize class-level adversarial learning by minimize classifier determinacy disparity while fixing the classifiers:

$$\min_{\theta_g} \beta L_{cdd}(x_i^t).$$

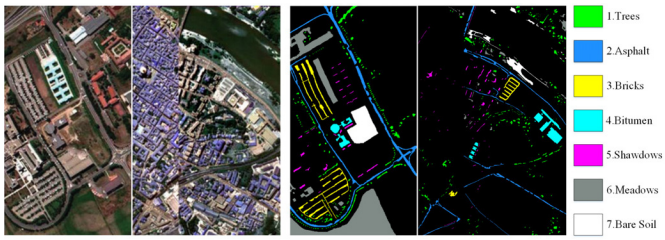
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III. Experiments and Result Analysis

In this section, the efficacy of the proposed UJADA is verified on three real-world hyperspectral data pairs representing domain adaptation across different geographic locations and time points. Among them, two pairs are widely used public datasets, and another was self-collected datasets that were captured by a UAV equipped with a hyperspectral camera under different lighting conditions within a day.

TABLE I
Number of Samples in Each Land Cover Category Within Pavia Data Pair

Class		Number of Samples	
ID	Name	PU	PC
1	Trees	3064	7598
2	Asphalt	6631	9248
3	Bricks	3682	2685
4	Bitumen	1330	7287
5	Shadows	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
Total		39332	39355



(a) (b) (c) (d)

Fig. 3. Hyperspectral images of Pavia data pair. (a) False-color images of PU. (b) False-color images of PC. (c) Ground truth of PU. (d) Ground truth of PC.

A. Data Description

1) Pavia Data Pair: The Pavia data pair includes two scenes: Pavia University (PU) and Pavia Center (PC). They were captured by the reflective optics spectrographic imaging system (ROSIS) sensor during a flight over an urban area of Pavia in Northern Italy, which contains high-resolution spectral information in the wavelength range from 0.43 to 0.86 after discarding noisy bands. The spatial resolution is 1.3 m. PU has a spatial resolution size of 610 × 340 and 103 bands, while the spatial dimension of PC is 1096 × 715 with 102 bands. In order to keep the consistent band number in the Pavia data pair, the last band of the PU original cube is abandoned. It is worth noting that the categories of the HSI images in two scenes are not exactly the same, and we select seven classes shared for experiments. Yang *et al.* [49] analyzed the tremendous challenge posed by the high intraclass variability and low interclass variability in the Pavia image classification. We list the number of samples in each class in Table I. The false-color images of the Pavia data pair are shown in Fig. 3(a) and (b), and the ground-truth images of them with seven shared classes are shown in Fig. 3(c) and (d), where black indicates the background.

2) Houston Data Pair: The Houston (HO) data pair was collected by the NSF-Funded Center for Airborne Laser Mapping (NCALM) over the campus area and the neighboring

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TABLE II
NUMBER OF SAMPLES IN EACH LAND COVER CATEGORY WITHIN HO DATA PAIR

Class		Number of Samples	
ID	Name	HO2013	HO2018
1	Healthy grass	1251	9799
2	Stressed grass	1254	32502
3	Trees	1244	13588
4	Water	325	266
5	Residential buildings	1268	39762
6	Commercial buildings	1244	223684
7	Road	1252	45810
	Total	7838	365411

urban area of the University of HO. The dataset consists of two multitemporal images that were acquired by ITRES-CASI 1500 Hyperspectral imager in 2012 and 2017. These two images were published in the 2013 and 2018 IEEE GRSS Data Fusion Contest. The spatial resolution was 2.5 and 1 m for HO 2013 and HO 2018, respectively. The HO2013 dataset consists of 349 × 1905 pixels with 144 bands, but the HO2018 dataset only has 48 bands with a spatial dimension of 601 × 2384. The spectral range of both images is 380–1050 nm, and every three bands in the HO2013 dataset were averaged to a new band so that its spectral dimensionality is consistent with HO2018 datasets [50]. Seven common categories were chosen from two images to conduct cross-scene classification. Liu *et al.* [15] and Zhang *et al.* [16] elaborated on the issue of spectral amplitude shift in the HO data pair. The number of samples in each class is listed in Table II. The false-color images of the HO data pair are presented in Fig. 4(a) and (b), and the ground-truth images of them with seven shared classes are shown in Fig. 4(c) and (d).

3) Playground Data Pair: To verify the practical value of the proposed method in the actual ultralow-altitude flight scenario, we built a scene containing eight types of ground objects on the campus playground at 10:00 A.M. and 4:00 P.M. on September 27, 2021, and the weather conditions in the morning and afternoon are cloudy and sunny, respectively. Then, we employed Hexacopter UAV equipped with the SPECIM_FX17E hyperspectral camera to implement ultralow-altitude flight aerial photography missions with atmospheric correction processing on a campus playground field, and the flight altitude was 30 m. We name the image taken in the morning as playground in the morning (PM) and the image captured in the afternoon as playground in the afternoon (PA). The spatial resolution of both two images in the playground data pair is 482 × 640, and the spectral wavelength of them ranges from 935 to 1720 nm with 224 bands. Similarly, the false-color images of the playground data pair are presented in Fig. 5(a) and (b), and the ground-truth images

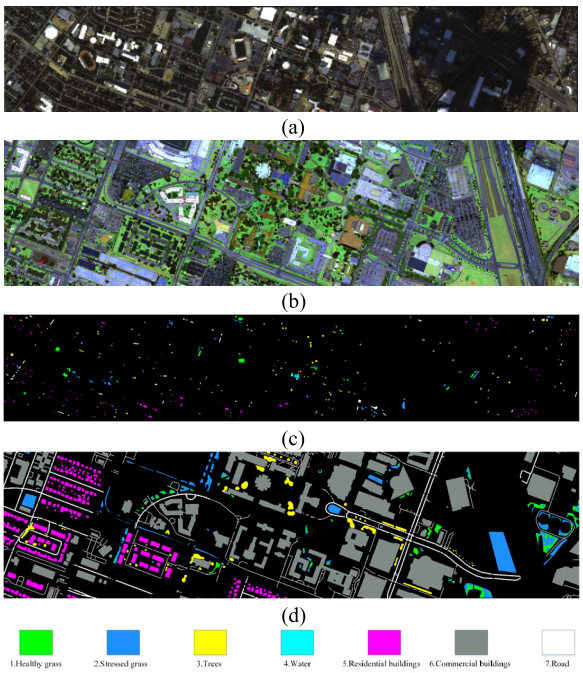


Fig. 4. Hyperspectral images of HO data pair. (a) False-color images of HO2013. (b) False-color images of HO2018. (c) Ground truth of HO2013. (d) Ground truth of HO2018.

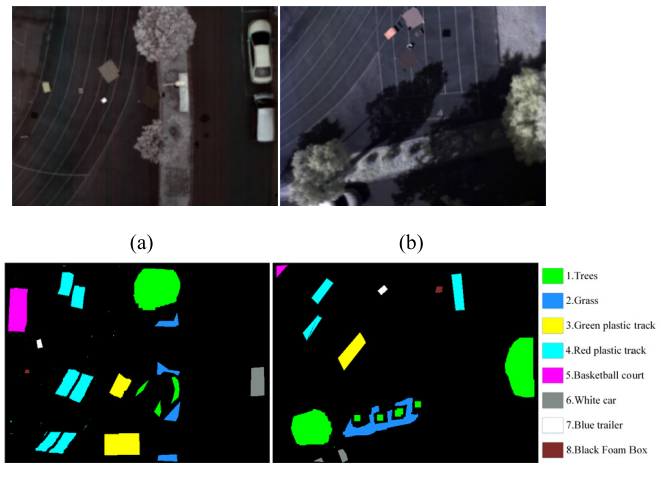


Fig. 5. Hyperspectral images of playground data pair. (a) False-color images of PM. (b) False-color images of PA. (c) Ground truth of PM. (d) Ground truth of PA.

of them with eight same classes are shown in Fig. 5(c) and (d). Also, we recorded the number of samples in each class in Table III. As shown in Fig. 6, under different illumination conditions in the cloudy morning and sunny afternoon, the shape of the spectral curve of each material is relatively stable, while the amplitude changes drastically, which means that the sensitivity to light intensity varies over different materials.

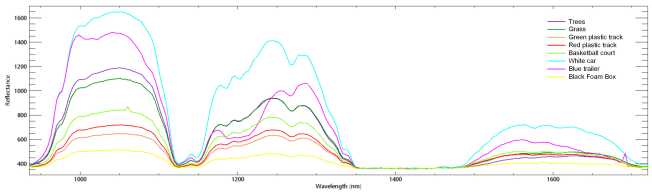
B. Implementation Details

In our experiments, the proposed method is compared with eight approaches in a series of experiments where two images in each data pair are taken as the source domain in turn. Regarding the Pavia and HO data pairs, we randomly

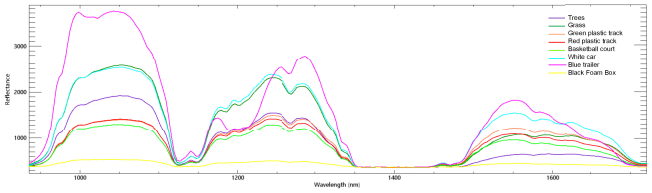
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TABLE III
NUMBER OF SAMPLES IN EACH LAND COVER CATEGORY WITHIN PLAYGROUND DATA PAIR

Class		Number of Samples	
ID	Name	PM	PA
1	Trees	10490	15962
2	Grass	3013	5177
3	Green plastic track	6068	2406
4	Red plastic track	9716	4191
5	Basketball court	4390	399
6	White car	2208	588
7	Blue trailer	233	288
8	Black Foam Box	89	226
	Total	36207	29237



(a)



(b)

Fig. 6. Spectral curve charts of playground data pair. (a) PM. (b) PA.

select 100 labeled samples from each class in the source domain as the source training set. Also, in each epoch, 700 unlabeled samples are randomly extracted from the target domain as the target training set, which is equal to the total number of source training samples. For the playground data pair, the number of samples for each class in the source training set and the total number of samples in the target training set are 80 and 640, respectively. Finally, we test the performance of each algorithm on all target data. The patch size of HSI cubes is set as 9×9, and the label of each cube is the category of the spatial geometric center pixel.

To compare the intradomain classification method and the cross-domain classification method, we utilized the FDSSC model [45] only trained on the source data to conduct transfer classification on the target data directly and define such experiments as "S-only," which does not employ any transfer strategy. The performance gap between "S-only" and other domain adaptation algorithms indicates the transferability of these methods. Besides, five typical domain adaptation methods, DANN [32], MCD [35], ADDA [33], JADA [44], and BCDM [42], were adopted to demonstrate the effectiveness of the JADA method in solving the HSI cross-domain classification problem and the superiority of adopting CDD [42] as a measure of classifier disparity during joint adversarial training. Among them, JADA is a combination of DANN and MCD, corresponding to domain-level alignment and class-level alignment, respectively. The proposed UJADA is a federated transfer strategy of DANN and BCDM, which adopts CDD as the discrepancy metric of biclassifier output probability. Therefore, comparative experiments support the ablation analysis. It is worth noting that the network structure of each component between different methods is consistent.

In addition, two state-of-the-art hyperspectral cross-domain classification algorithms, UDAD [40] and UDACA [41], are also introduced for cross-domain classification performance comparison on the three data pairs.

All experiments were implemented on an Nvidia TITAN RTX with 24-GB RAM by using the PyTorch framework and python 3.6 simulation platform. We trained the network for 200 epochs with a mini-batch size of 32, and the Adam optimizer was adopted for backpropagation in the training phase with a learning rate of 0.00005. In addition, we noticed that UJADA performs stably under different tradeoff hyperparameter settings by deep embedded validation (DEV) [51], and thus, it was fixed as 0.01 throughout all execution in this article. The average accuracy (AA), overall accuracy (OA), and statistic kappa coefficient (Kappa) are applied as objective criteria to comprehensively evaluate the competing methods. All classification results reported in this article are averaged by five-time conduction.

C. Results and Ablation Analysis

For the Pavia data pair, we construct an annotated source training set on the PU dataset and perform unsupervised transfer and classification tests on the PC scene. The colored classification map is shown in Fig. 7, and the classification results are given in Table IV. Then, the second experiment swaps the domain identities of PU and PC images and the unsupervised classification outcomes of PU image are presented in Fig. 8 and Table V. Compared with other domain adaptation methods, our algorithm UJADA achieves the best classification performance on PC images. The JADA approach yields the better results than both MCD and DANN methods. The classification performance of BCDM significantly outperforms JADA, ADDA, MCD, and DANN. In addition, UJADA also provides the optimal result on PU image and reaches the highest classification accuracy on most categories. BCDM still outperforms JADA, both exceeding DANN and MCD models.

As can be observed, all domain adaptation methods have a certain performance improvement compared to the S-only method and JADA outperforms both MCD and DANN, which demonstrates that JADA is effective for the HSICC task and can be attributed to that joint adversarial alignment enables the domain-level adversarial module and the class-level adversarial module to complement each other in a unified training process, thus achieving remarkable adaptive

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TABLE IV
CROSS-SCENE CLASSIFICATION RESULTS WITH DIFFERENT METHODS OF PU → PC

Class	S-only	MCD	DANN	ADDA	JADA	BCDM	UADA	UDACA	UJADA
1	0.902	0.946	0.932	0.866	0.947	0.960	0.877	0.914	**0.979**
2	0.484	0.964	0.655	0.792	0.946	**0.998**	0.942	0.992	0.993
3	0.577	0.605	**1.000**	0.896	**1.000**	0.776	0.852	0.962	0.975
4	0.242	0.020	0.489	0.715	0.813	0.846	**0.936**	0.842	0.838
5	0.853	0.979	0.985	0.995	0.928	0.999	0.996	**1.000**	**1.000**
6	0.311	0.908	0.899	**0.957**	0.797	0.920	0.712	0.901	0.916
7	0.385	0.857	0.783	0.795	0.663	**0.964**	0.846	0.945	0.905
AA	0.537	0.754	0.820	0.860	0.871	0.923	0.880	0.936	**0.944**
OA	0.523	0.740	0.766	0.828	0.865	0.936	0.880	0.929	**0.940**
Kappa	0.441	0.692	0.726	0.795	0.838	0.923	0.855	0.914	**0.928**

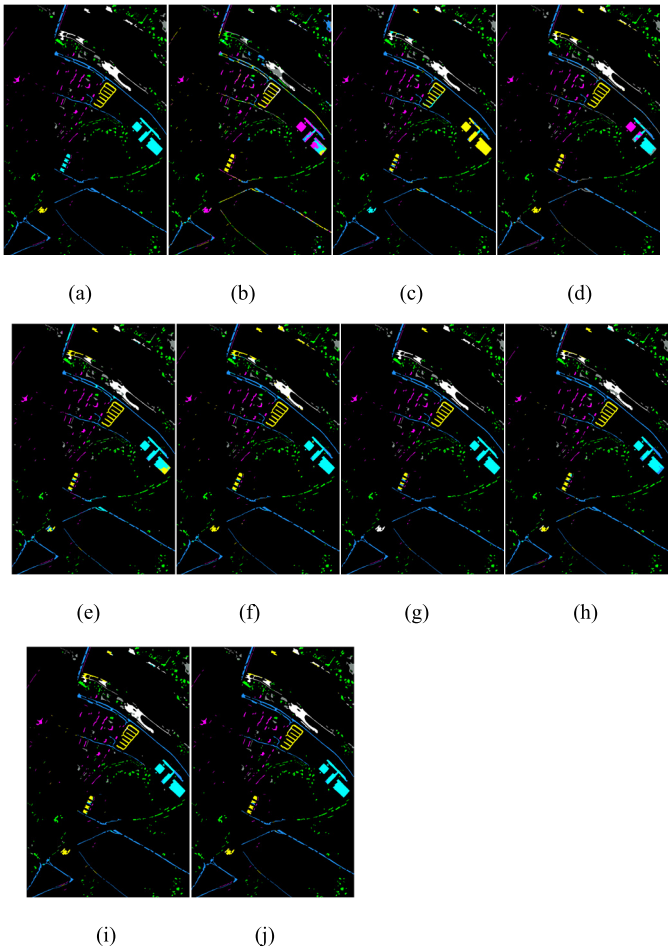


Fig. 7. Cross-scene classification maps with all the compared methods in PU → PC. (a) Ground-truth image. (b) S-only. (c) MCD. (d) DANN. (e) ADDA. (f) JADA. (g) BCDM. (h) UADA. (i) UDACA. (j) UJADA.

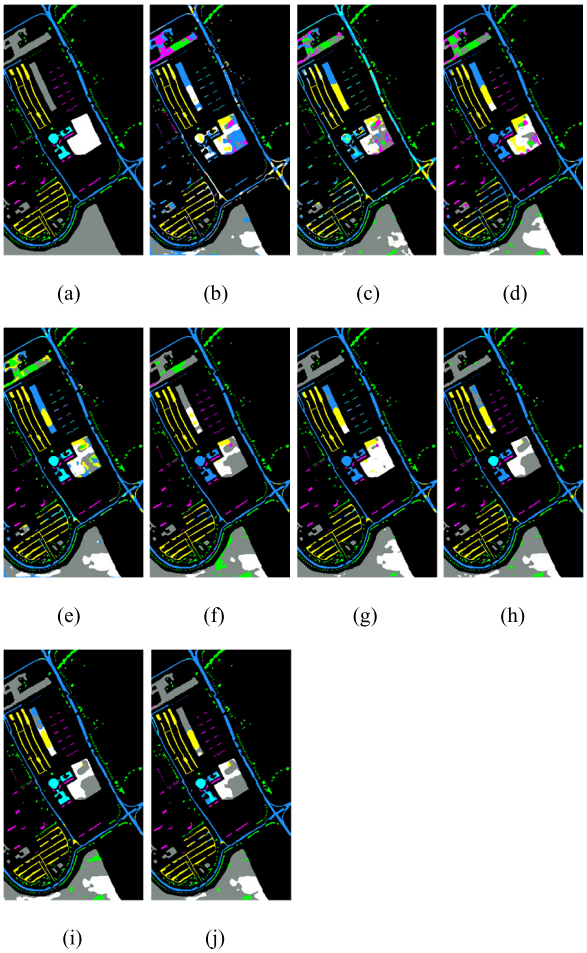


Fig. 8. Cross-scene classification maps with all the compared methods in PC → PU. (a) Ground-truth image. (b) S-only. (c) MCD. (d) DANN. (e) ADDA. (f) JADA. (g) BCDM. (h) UADA. (i) UDACA. (j) UJADA.

capabilities. Notably, the BCDM approach achieves amazing classification performance, which verifies that the CDD metric plays a key role in measuring the difference between two task-specific classifiers at a more fine-grained level. Besides, our UJADA method surpasses two state-of-the-art hyperspectral UDA methods in terms of three evaluation indicators.

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TABLE V
CROSS-SCENE CLASSIFICATION RESULTS WITH DIFFERENT METHODS OF PC → PU

Class	S-only	MCD	DANN	ADDA	JADA	BCDM	UADA	UDACA	UJADA
1	0.392	0.929	0.917	0.809	0.959	**0.969**	0.927	0.776	0.901
2	0.789	0.437	**0.992**	0.879	0.977	0.967	0.792	0.829	0.975
3	0.821	0.613	0.763	0.899	0.910	0.939	0.920	0.902	**0.949**
4	0.000	0.396	0.000	0.806	0.000	0.000	0.462	**0.869**	0.624
5	0.000	0.035	0.965	0.798	1.000	0.806	0.899	0.977	**1.000**
6	0.536	0.677	0.530	0.522	0.711	0.694	0.792	**0.799**	0.770
7	0.334	0.344	0.514	0.391	0.351	**0.857**	0.479	0.502	0.457
AA	0.410	0.490	0.669	0.729	0.701	0.747	0.753	0.808	**0.811**
OA	0.537	0.583	0.650	0.639	0.730	0.784	0.766	0.780	**0.792**
Kappa	0.400	0.444	0.554	0.538	0.633	0.714	0.678	0.698	**0.716**

TABLE VI
CROSS-SCENE CLASSIFICATION RESULTS WITH DIFFERENT METHODS OF HO 2013 → HO 2018

Class	S-only	MCD	DANN	ADDA	JADA	BCDM	UADA	UDACA	UJADA
1	0.381	0.432	0.671	0.667	0.794	0.770	0.828	0.752	**0.855**
2	0.534	**0.951**	0.917	0.815	0.830	0.833	0.840	0.920	0.858
3	0.659	0.936	0.968	**0.997**	0.964	0.985	0.965	0.978	0.980
4	0.237	0.391	**1.000**	0.996	0.312	0.977	**1.000**	**1.000**	**1.000**
5	0.131	0.834	0.864	0.793	0.668	**0.907**	0.893	0.886	0.853
6	0.781	0.652	0.642	0.742	**0.783**	0.709	0.673	0.675	0.729
7	0.622	0.697	0.799	0.655	0.648	0.730	0.803	0.826	**0.829**
AA	0.478	0.699	0.838	0.809	0.714	0.845	0.857	0.862	**0.872**
OA	0.653	0.708	0.724	0.751	0.764	0.756	0.743	0.752	**0.779**
Kappa	0.425	0.575	0.602	0.617	0.627	0.635	0.626	0.641	**0.667**

For the HO data pair, the cross-domain classification maps and the unsupervised classification results of all the compared algorithms on the case HO 2013 → HO 2018 are shown in Fig. 9 and Table VI, respectively. Obviously, due to the extremely imbalanced sample size of each category on the HO2018 dataset, the classification results of different methods among three indicators are inconsistent, which also appears in the classification results of the PU scene. Despite this, our approach produces the best performance. Then, we train the HO 2018 dataset as the source domain and test the unsupervised classification performance on HO 2013 scene; the experimental results are shown in Fig. 10 and Table VII. Similarly, the proposed UJADA method still provides the best classification result. This reconfirms the conclusions drawn on the Pavia data pair compared to other domain adaptation algorithms. Besides, it is worth noting that the overall classification performance of all methods on the HO2013 image is better than that of the HO2018 scene. An analogous conclusion can be drawn in the Pavia data pair when training the PC scene as the source domain, which can be explained by the more visible scattered distribution of samples and the large spatial variability of spectral features within categories. Moreover, taking 3-D cubes as input samples considers redundant information during feature learning.

The last set of experiments is conducted in playground data pair with high spatial resolution and only amplitude shift exists, which means the few domain discrepancies in the HSICC task. In the case of PM → PA, the colorful classification maps and values of three classification indexes for different methods are given in Fig. 11 and Table VIII, respectively. Our method yields the optimal experimental result. Furthermore, we also perform UDA experiments with the source

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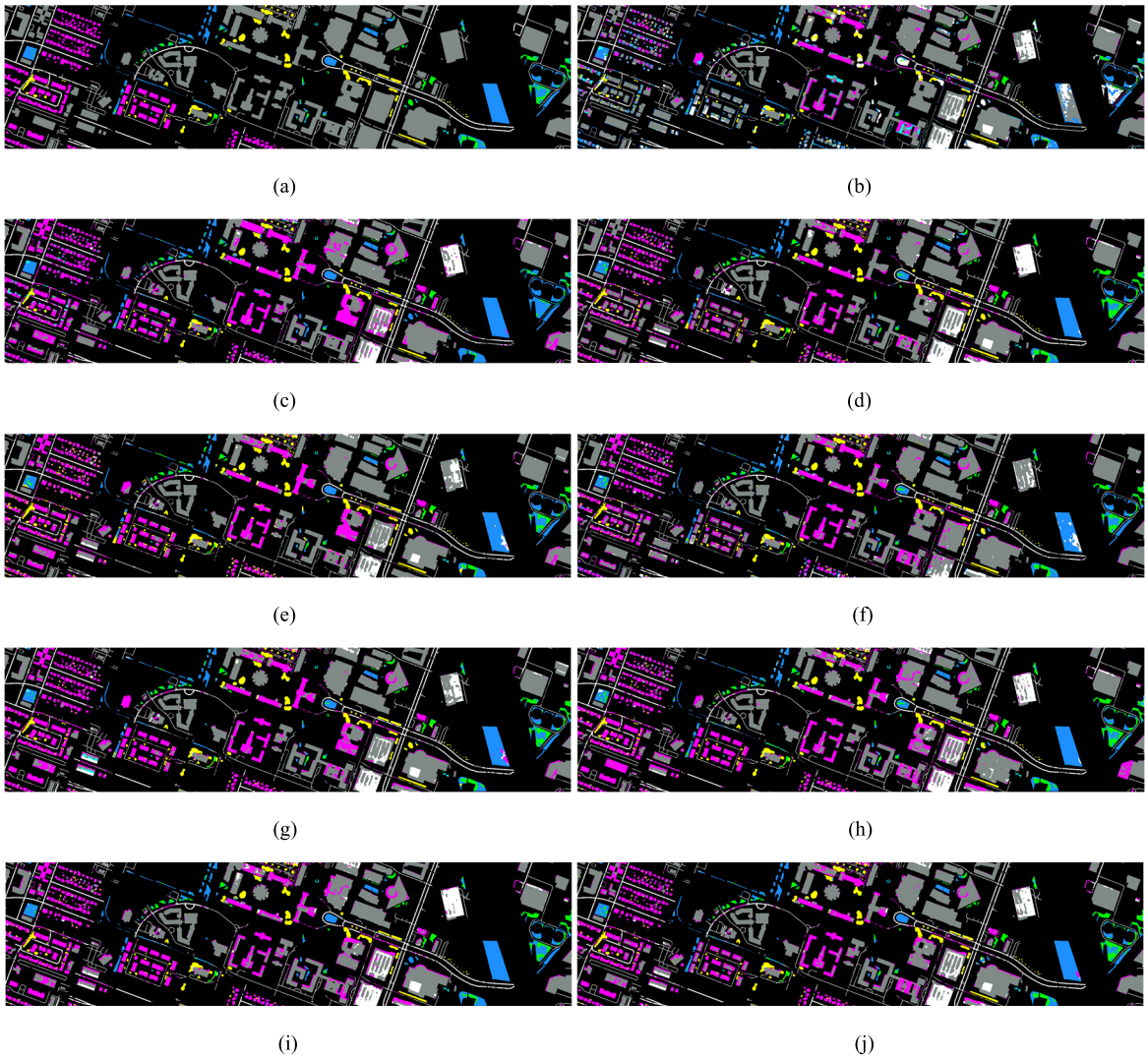


Fig. 9. Cross-scene classification maps with all the compared methods in HO 2013 → HO 2018. (a) Ground-truth image. (b) S-only. (c) MCD. (d) DANN. (e) ADDA. (f) JADA. (g) BCDM. (h) UADA. (i) UDACA. (j) UJADA.

TABLE VII
CROSS-SCENE CLASSIFICATION RESULTS WITH DIFFERENT METHODS OF HO 2018 → HO 2013

Class	S-only	MCD	DANN	ADDA	JADA	BCDM	UADA	UDACA	UJADA
1	**0.858**	0.775	0.841	0.799	0.851	0.795	0.850	0.856	0.819
2	0.734	0.849	0.833	**0.874**	0.864	0.868	0.833	0.869	0.837
3	0875	0.893	**0.944**	0.932	0.940	0.937	0.928	0.943	0.928
4	0.511	0.812	0.874	**0.877**	**0.877**	**0.877**	0.875	0.871	0.874
5	0.363	0.621	0.760	0.621	0.617	0.685	0.773	**0.801**	**0.870**
6	0.616	0.370	0.493	0.639	0.677	**0.853**	0.575	0.677	0.829
7	0.907	0.907	**0.977**	0.927	0.972	0.875	0.966	0.945	0.917
AA	0.695	0.747	0.817	0.810	0.828	0.841	0.828	0.851	**0.867**
OA	0.715	0.739	0.811	0.802	0.822	0.837	0.823	0.849	**0.867**
Kappa	0.663	0.693	0.777	0.765	0.789	0.806	0.793	0.818	**0.842**

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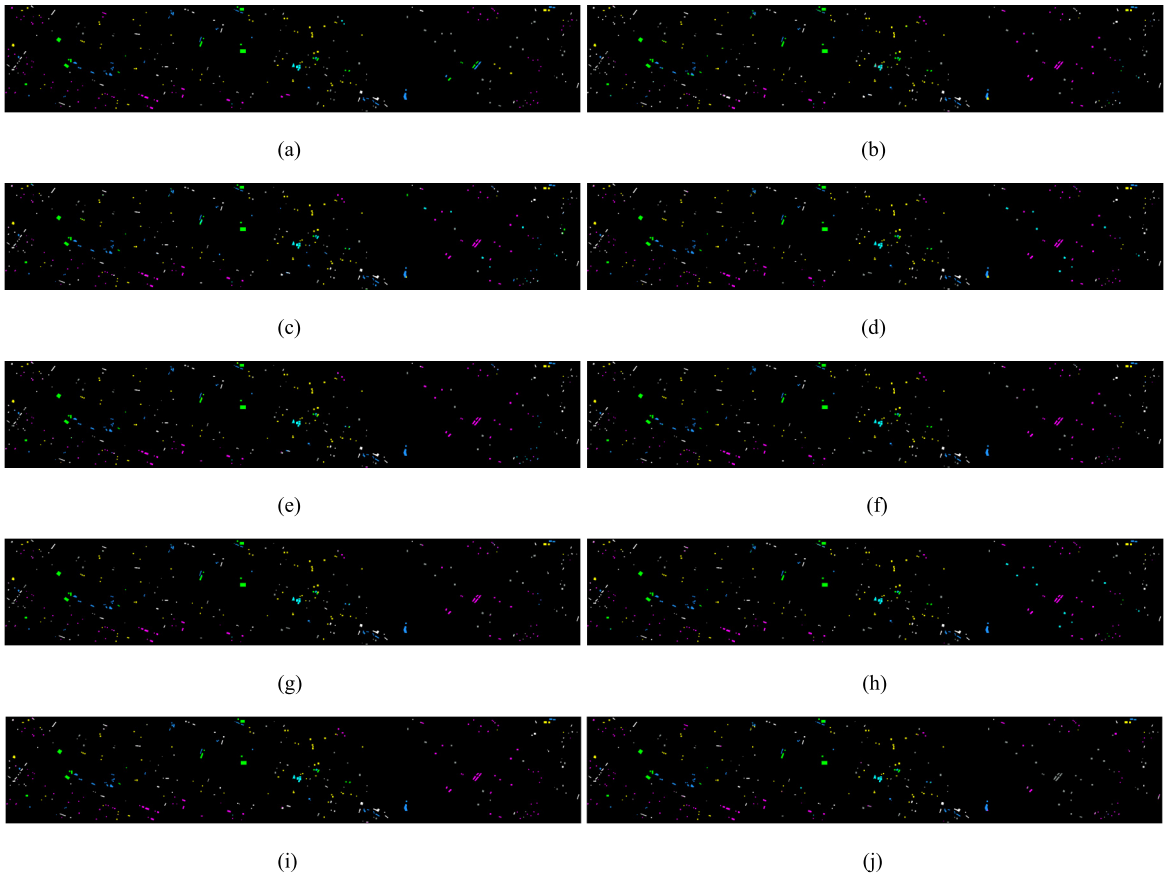


Fig. 10. Cross-scene classification maps with all the compared methods in HO 2018 → HO 2013. (a) Ground-truth image. (b) S-only. (c) MCD. (d) DANN. (e) ADDA. (f) JADA. (g) BCDM. (h) UADA. (i) UDACA. (j) UJADA.

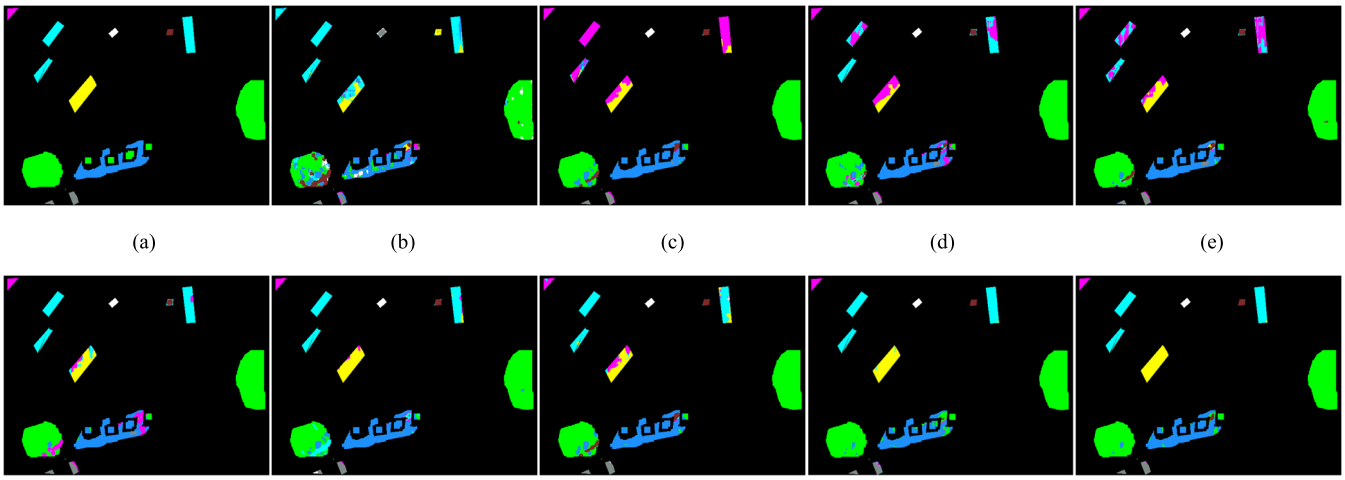


Fig. 11. Cross-scene classification maps with all the compared methods in PM → PA. (a) Ground-truth image. (b) S-only. (c) MCD. (d) DANN. (e) ADDA. (f) JADA. (g) BCDM. (h) UADA. (i) UDACA. (j) UJADA.

and target domains of the playground data pair exchanged, and the classification results are presented in Fig. 12 and Table IX. Similarly, our UJADA model also achieves the best performance. These are close to the achievable performance by supervised training. As can be observed, both joint adversarial alignment and the CDD metric play a positive role in terms of improving the adaptive ability of the proposed model. In addition, the overall classification performance of the DANN method in each scene is better than that of the MCD method. This can be interpreted as the characteristics of the cross-scene data described in Section III-A, which causes the difference in domain distribution between hyperspectral

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TABLE VIII
CROSS-SCENE CLASSIFICATION RESULTS WITH DIFFERENT METHODS OF PM → PA

Class	S-only	MCD	DANN	ADDA	JADA	BCDM	UADA	UDACA	UJADA
1	0.724	0.882	0.870	0.909	0.903	0.852	0.888	0.941	**0.943**
2	0.763	**0.946**	0.838	0.897	0.824	0.926	0.905	0.856	0.913
3	0.419	0.588	0.416	0.697	0.722	0.948	0.776	0.973	**1.000**
4	0.913	0.028	0.639	0.431	0.974	0.951	0.906	0.997	**0.997**
5	0.003	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.930	**1.000**	**1.000**
6	0.682	0.866	0.886	0.883	0.825	0.869	0.973	0.871	**0.985**
7	0.219	0.997	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
8	0.035	0.978	0.792	0.832	0.712	0.956	0.987	0.989	**0.996**
AA	0.470	0.785	0.805	0.831	0.870	0.938	0.920	0.953	**0.979**
OA	0.712	0.749	0.797	0.822	0.883	0.892	0.889	0.937	**0.953**
Kappa	0.588	0.636	0.703	0.737	0.826	0.840	0.839	0.906	**0.927**

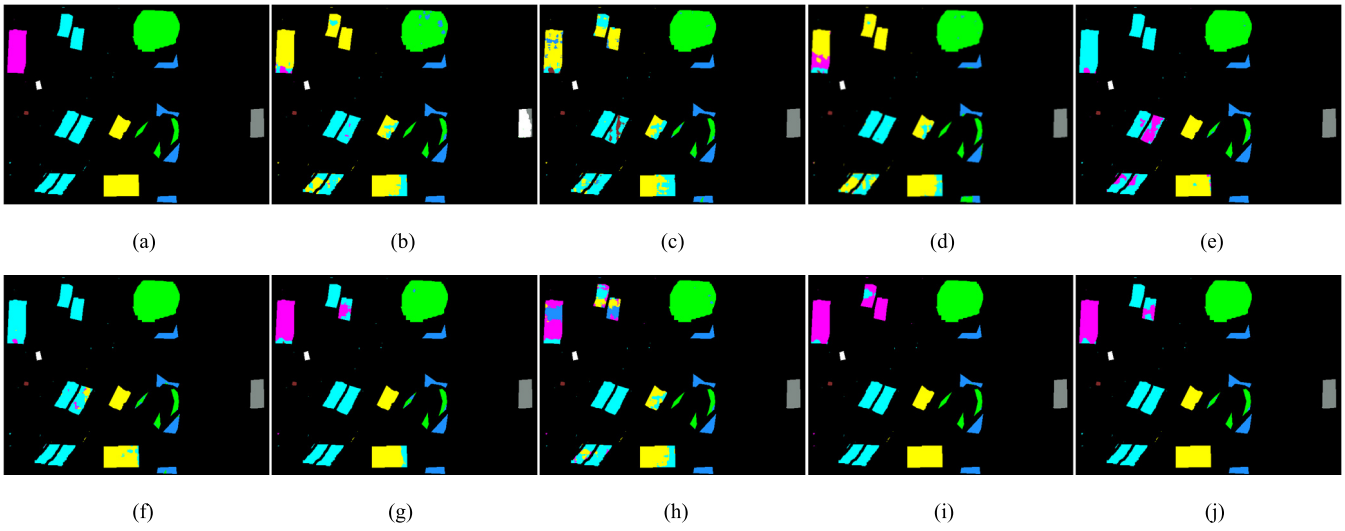


Fig. 12. Cross-scene classification maps with all the compared methods in PA → PM. (a) Ground-truth image. (b) S-only. (c) MCD. (d) DANN. (e) ADDA. (f) JADA. (g) BCDM. (h) UADA. (i) UDACA. (j) UJADA.

images to be more apparent than the difference in category distribution. This also illustrates that our UJADA algorithm has tremendous practical value and application potential in UAV ultralow-altitude remote sensing missions under different illumination conditions.

D. Analysis of the Parameter Effect

In this section, we analyze the effect of two variables on the performance of the proposed algorithm, namely, the patch size and the number of training samples in the source domain. The impact of the number of training samples in the source domain is presented in the line charts in Fig. 13. Regarding the Pavia and the HO data pairs, we conducted comparative experiments with training sample sizes of 20, 40, 60, 80, and 100 for each category of the source domain. As for the playground data pair, we compared the classification performance with source training sample sizes of 20, 40, 60, and 80 for each category. It can be observed that the more training samples selected from the source domain will obtain the better classification performance of AA, OA, and Kappa. Also, our method outperforms the classification results of existing methods [40], [52], and the latter proposes an iterative training sample updating (ITSU) approach.

In addition, cross-scene classification results under different patch sizes toward six scenarios are shown in Fig. 14. As shown in Fig. 14(a), our UJADA model performs the highest AA with the size set as 9×9 for PC (0.944), HO2018 (0.872), and HO2013 (0.867). Also, when the size is set to 11×11, PU, PA, and PM achieve the best AA of 0.868, 0.987, and 0.995, respectively.

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TABLE IX
CROSS-SCENE CLASSIFICATION RESULTS WITH DIFFERENT METHODS OF PA → PM

Class	S-only	MCD	DANN	ADDA	JADA	BCDM	UADA	UDACA	UJADA
1	0.947	0.985	0.989	0.988	**0.993**	0.982	0.989	0.991	0.992
2	**1.000**	0.982	0.835	**1.000**	0.977	**1.000**	**1.000**	0.997	**1.000**
3	0.766	0.610	0.812	0.971	0.941	0.900	0.754	**1.000**	**1.000**
4	0.599	0.669	0.543	0.777	**0.947**	0.925	0.711	0.730	0.934
5	0.069	0.002	0.285	0.041	0.026	0.918	0.614	0.943	**0.954**
6	0.180	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
7	**1.000**	**1.000**	0.996	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
8	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
AA	0.695	0.781	0.807	0.847	0.860	0.966	0.883	0.958	**0.985**
OA	0.675	0.719	0.742	0.816	0.854	0.948	0.831	0.917	**0.974**
Kappa	0.592	0.645	0.675	0.764	0.811	0.934	0.788	0.901	**0.968**

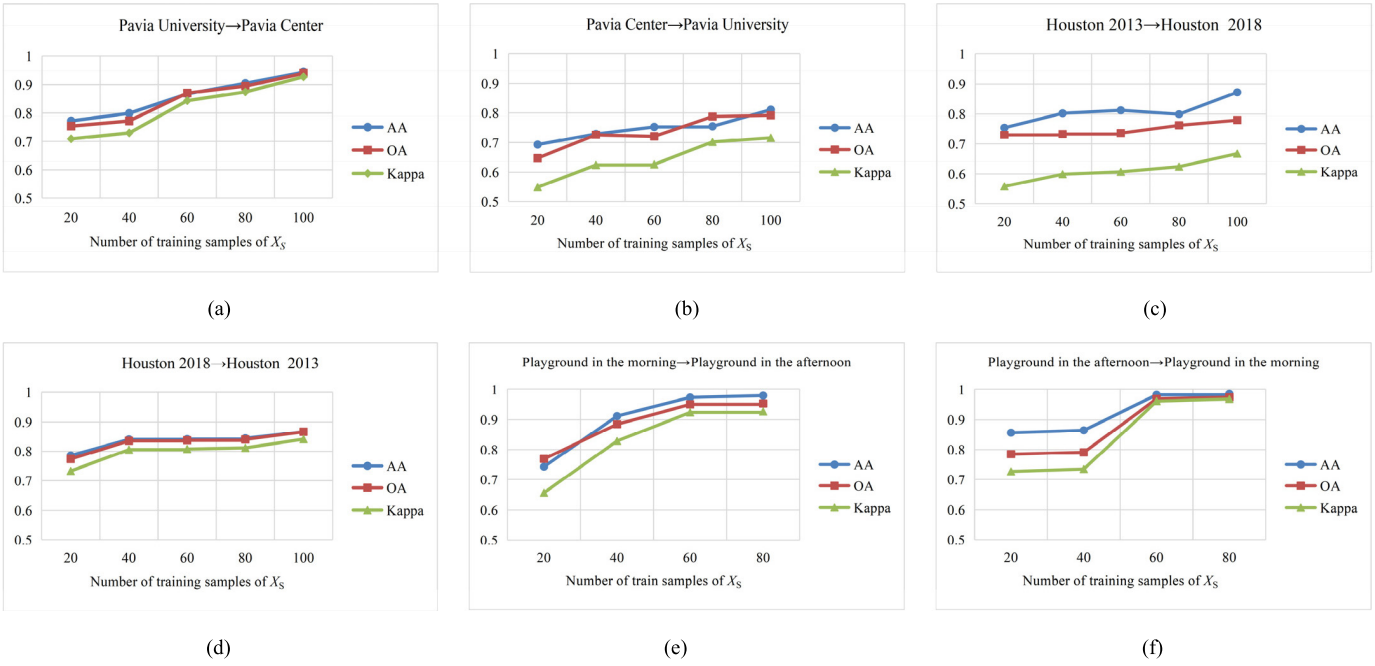
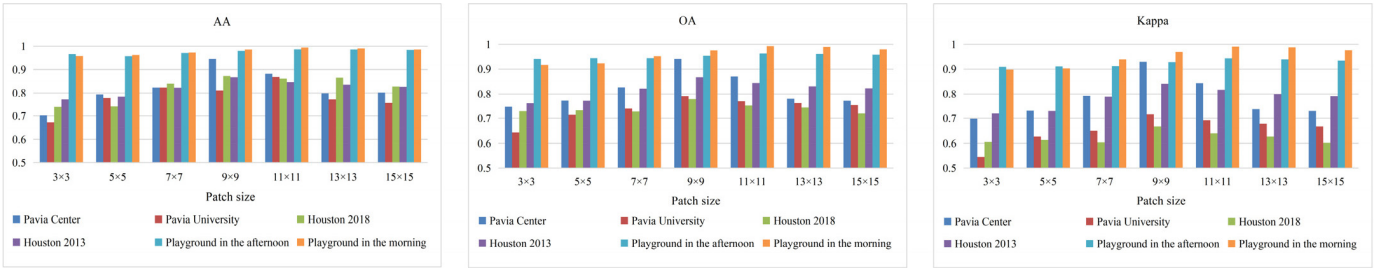


Fig. 13. Classification performance achieved by the UJADA with increasing training samples from the source domain. (a) PU → PC. (b) PC → PU. (c) HO 2013→ HO 2018. (d) HO 2018 → HO 2013. (e) PM → PA. (f) PA → PM.

Furthermore, the histograms in Fig. 14(b) and (c), respectively, reflect the variation of OA and Kappa at different patch sizes. As can be observed, both the Pavia data pair and HO data pair obtain the highest OA and Kappa with the size of 9 × 9, but our model yields the best OA and Kappa for playground data pair with the patch size set as 11 × 11. Specifically, the best OA for classification on PC, PU, HO2018, HO2013, PA, and PM scenes are 0.940, 0.792, 0.779, 0.867, 0.962, and 0.993, respectively. Also, the optimal Kappa values are 0.928, 0.716, 0.667, 0.842, 0.942, and 0.991.

In particular, with the increase of patch size, the classification performance among three indicators reflects a tendency of rising first and then falling slowly after reaching the peak value. Besides, the patch size has less impact on the classification performance of the proposed method on the playground data pair than on the two public data pairs, which can be attributed to the high spatial resolution of the playground data pair. This also demonstrates the excellent effectiveness of our model in ultralow-altitude hyperspectral remote sensing applications. All in all, we summarize that both the patch

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(a) (b) (c)

Fig. 14. Impacts of different patch sizes on the cross-scene classification performance of UJADA on six HSI scenarios. (a) AA. (b) OA. (c) Kappa.

size and the number of training samples affect interdomain adaptability.

IV. Conclusion

In this article, we explore the issue of hyperspectral cross-scene classification and analyze two major challenges it faces, including the formidable work of obtaining labeled samples and the phenomenon of spectral shift between different scenes. Aim at this, a UJADA framework is proposed to decrease the interdomain disparity, enabling efficient unsupervised classification in target hyperspectral images. The proposed UJADA method integrates domain-level adversarial alignment and class-level adversarial alignment, which can complement each other in an end-to-end training process. Moreover, we adopt the CDD metric to finely measure the discrepancy of biclassifier output probability, ensuring that the learned transferable features are discriminative and finally achieve better domain adaptation ability. Besides, our UJADA model takes advantage of a dense-based spectral–spatial convolution network as the feature extractor to comprehensively consider the spatial and spectral information of hyperspectral images during feature learning, which promotes the distinguishability of generated features. The superiority of the proposed UJADA method over several other methods is verified by experiments on public datasets and self-collected ultralow-altitude hyperspectral images. Of particular note, our UJADA method overcomes the spectral shift problem of real-world hyperspectral images under different illumination conditions.

In the future, our goal is to probe hyperspectral cross-scene classification problems under larger domain shift, such as sunny and rainy weather. Furthermore, we will delve into the characteristics of hyperspectral data and the mechanism of advanced domain adaptation algorithms to develop efficient hyperspectral image cross-domain classification approaches with deep networks.

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